SIXTH SEMESTER

Course Code	Course Name	Th	Tu	Pr	Credits	Evaluation		
						CA	ESE	Total
23CB602T	Information Security	3	-	-	3	30	70	100

Unit I Overview of Security Parameters

Overview of Security Parameters: Confidentiality, integrity and availability; Security violation and threats; Security policy and procedure; Assumptions and Trust; Security Assurance, Implementation and Operational Issues; Security Life Cycle.

1) Give two examples of Confidentiality breach.
2) Explain how encryption helps in maintaining confidentiality.
3) Describe threats affecting availability.
4) Explain CIA triad in banking system.
5) Discuss how CIA principles are applied in an e-commerce system.
6) Explain Security Violation and its types with the help of example.
7) Differentiate between passive and active attacks.
8) Differentiate between threat, vulnerability, and attack.
9) Explain passive and active attacks with examples.
10) Discuss different types of security threats with examples.
11) Explain security violations and their impact on CIA triad.
12) Differentiate between Security Policy and Security Procedure.
13) Write short note on Incident Response Policy.
14) Explain importance of security policy in an organization.
15) What is Trusted Computing Base (TCB)?
16) Explain the concept of Trust and Assumptions in system security.
17) What is Trusted Computing Base? Explain its components.
18) Discuss the importance of trust and assumptions in designing secure systems.
19) Define Security Assurance. What is vulnerability assessment, Define risk assessment.
20) What is patch management, Define auditing.
21) Explain Security Assurance with examples.
22) What is difference between vulnerability assessment and penetration testing?
23) Explain importance of regular security audits.
24) Explain challenges in implementing security controls.
25) Describe operational issues in maintaining information security.
26) Define Security Life Cycle. List and explain phases of Security Life Cycle.
27) What is incident response and response handling.
28) Write short note on incident handling.
29) Explain importance of continuous improvement in security.
30) Apply the Security Wheel model to design security policies for an E-commerce system.

Unit II Access Control Models & Security policies

Access Control Models: Discretionary, mandatory, roll-based and task-based models, unified models, access control algebra, temporal and spatio-temporal models. **Security Policies:** Confidentiality policies, integrity policies, hybrid policies, non-interference and policy composition, international standards. Information Rights management.

1) Explain Discretionary Access Control (DAC) model with diagram. Give suitable example of DAC system.
2) Describe Access Control Matrix representation. What is an Access Control List (ACL) and capability list.
3) Differentiate DAC and MAC.
4) Explain Discretionary Access Control model with Access Control Matrix diagram.
5) Define Mandatory Access Control (MAC). Explain MAC with example.
6) What are characteristics of MAC?
7) Compare MAC and RBAC models.
8) Explain RBAC model with diagram. Describe types of RBAC.
9) Compare RBAC with DAC and MAC.
10) Explain TBAC model with example. Advantages of TBAC in workflow systems.
11) Difference between RBAC and TBAC.
12) Explain Task-Based Access Control with workflow diagram.
13) Discuss the role of TBAC in healthcare systems.
14) Compare TBAC and RBAC in enterprise systems.
15) Define hybrid access control.
16) What is policy integration?
17) Explain unified model combining DAC, MAC and RBAC.
18) Challenges in implementing unified model.
19) Explain unified access control model with architecture diagram.
20) Explain Access Control Algebra with example.
21) Relation between Access Control Matrix and Algebra
22) Discuss how algebra helps in policy verification.
23) Derive access control expression using algebraic operations.
24) Define Temporal Access Control.
25) Explain Spatio-temporal model with example. Applications of spatio-temporal models.
26) Explain confidentiality policies in detail with Bell-LaPadula model.
27) Discuss information flow control mechanisms.
28) Compare confidentiality and integrity policies.
29) Explain non-interference model in detail.
30) Explain hybrid security policies.
31) Explain policy composition challenges.
32) Explain non-interference model in detail.
33) Discuss policy composition and conflict resolution.
34) What is ISO and its different types with suitable example.
35) Explain international standards in information security in detail.
36) Define Information Rights Management. Evaluate challenges in implementing IRM systems.
37) What is digital rights management? Advantages of IRM.
38) Explain Information Rights Management with example.
39) Difference between DRM and IRM.